Low-Voltage 6T SRAM Reliability

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Mission

To design and simulate a low-voltage 6T SRAM cell and investigate how reducing supply voltage affects data reliability, power consumption, and speed, while exploring transistor sizing to improve performance.

Mission change: Our original focus was understanding SRAM reliability. We have now narrowed it to experimentally testing voltage scaling and possible improvements through transistor sizing.

User Story

Semiconductor Manufacturer

- Low power 6T SRAM cell so that I can reduce power consumption and manufacturing costs while maintaining performance and stability in integrated circuits.

VLSI Researchers

- I want to analyze how reducing supply voltage (VDD) affects the stability and performance of a 6T SRAM cell so that I can identify the minimum reliable operating voltage and explore.

General Electronics Consumers

- Devices to use energy-efficient and reliable SRAM so that they can have longer battery life, maintain fast performance, and operate without memory-related errors.

Tools

Cadence Virtuoso: Build and simulate the transistor-level 6T SRAM circuit.

SPICE / Cadence Spectre: Analyze transistor behavior at different supply voltages.

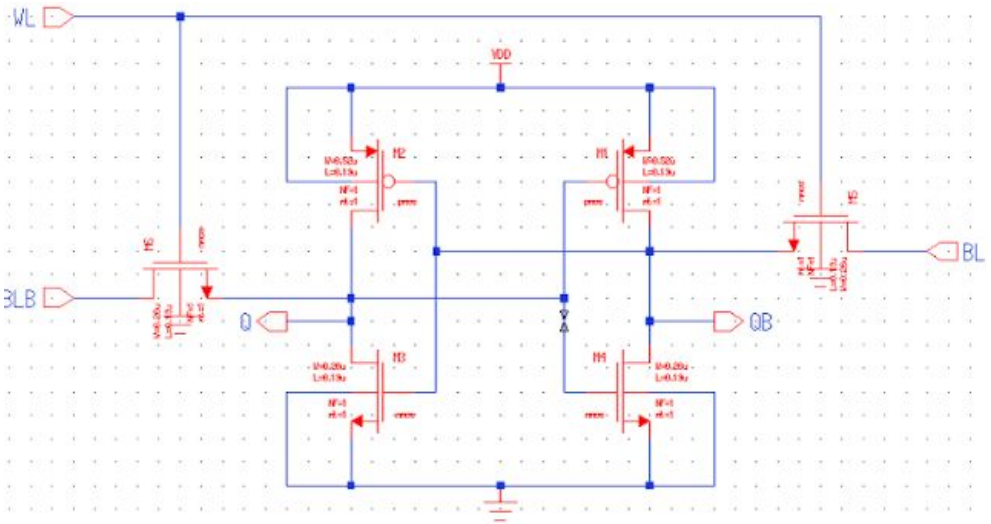
Waveform Viewer: Observe stored-node voltages, read/write transitions, and delays.

Research Papers: Understand SRAM stability and identify useful experimental methods.

Progress

1. Initial scripting

2. Basic PSpice design
   a. Simple 6 transistor setup of SRAM
   b. Baseline design for modifications



What we have learned

Main takeaway: Lower supply voltage does not simply mean lower power. It also affects the reliability and speed of SRAM.

- Read stability and write ability can require conflicting transistor-sizing choices.

- The SRAM cell may successfully hold data but still experience read or write failures.

- There is no single ideal voltage; reliable operation depends on the transistor models, sizing, and operating conditions.

What surprised us: Making certain transistors stronger can improve read stability while making writing more difficult.

We have learned this from the literature but have not yet tested it in our own simulations.

Sprint 2 Goals

- Have the initial chip tested under basic conditions to ensure it can read and write accurately
- Devise a list of parameters to change and their values. Ex:
  - Sizing
  - Layout
  - Wiring
  - Materials
  - Gate length
  - Temperature